

FULL-STACK WEB SOLUTIONS DEVELOPMENT & UI ENGINEERING WITH AI INTEGRATION

Develops front-end and full-stack web engineers with UI/UX craft: responsive, accessible interfaces in React (with Next.js) and Angular using TypeScript, modern state management, and design-system practice, extending to full-stack work with Node.js, Nest.js, GraphQL, and databases. A dedicated UI/UX strand bridges design and implementation, and a JavaScript-native generative-AI strand (LLM integration, RAG, agents/MCP, AI-assisted coding) enables intelligent web features. Graduates suit Front-End Developer, UI Engineer, and Full-Stack JavaScript Developer roles, with AI web-integration as a secondary capability. Produces AI-assisted web engineers, not machine-learning engineers.

JOB PROFILES

Full-Stack JavaScript Developer (MERN / MEAN / MEVN) ·
Frontend Developer (React / Angular / Vue & TypeScript) · UI/UX Developer (UX Engineer) ·
AI Application Developer (JS/TS) - (Generative AI — LLM · RAG · Agentic) ·
AI Automation Developer (n8n)



TOTAL HOURS

1,408^{hrs}

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



Where graduates work

_VOIS

Link

Fawry

EVA Pharma

Yodawi

Souq

Etisalat Misr

Aman

Sprinters

Genwin

Booking

EventBrite

Vezeeta

Valeo

Etisalat



Who may apply

All University Graduates



Selection process

01 IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview



Interview with the technical committee.

04 Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.

Program content – 1,408 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Software Development Fundamentals	369 hrs	26%	Database Fundamentals, SOLID & Design Patterns, Software Engineering & UML, Relational Databases, JS Unit Testing & TDD, Data Structures & Algorithms, OOP, Docker & Kubernetes, MS Azure, CI/CD, Source Control, Secure Coding, IIS Administration, operating Systems, Networks, Agile, E-Business, Digital Marketing
● Core	670 hrs	48%	Client-Side Development, Responsive Design Frameworks , Responsive Design & CSS Pre, JS Packaging & Bundling, Advanced JS, ES6 & TS, PWA, JS Frameworks & Architecture, React with AI, Next with AI, Vue with AI, Nuxt with AI, Angular with AI, Angular NgRx & RxJS, Node.js with AI, NoSQL, Nest, GraphQL, Material UI with Ai, UI/UX Design & Wireframing, Design Thinking, Visual Design, Photoshop, Illustrator, UX Writing, Design & Prototyping, Gen AI & Prompt Engineering, AI-Assisted Coding & AI Agents, AI Automation & Orchestration, Gen-AI Development Foundations, RAG & Agentic AI
● Soft Skills & Career Development	69 hrs	5%	Communication Skills, Presentation Skills, Job Seeking Skills, Professional Demeanor, Remote Working Practices, Teamwork & Leadership Skills, Freelancing
● Graduation Project	300 hrs	21%	

75% face to face Learning| 25 % Online, Common Hardware, Common Software

Student Deliverable

Each student must deliver at least ONE freelancing job and an international certificate based on his track

Before you apply

These topics will be discussed with you in the interviews (Resources applicants can visit or study before interview)

[An Introduction to Programming could be found at Coursera](https://www.coursera.org/learn/intro-programming)
https://www.coursera.org/learn/intro-programming

[Computer Programming Tutorial at](https://www.tutorialspoint.com/computer_programming/index.htm)
https://www.tutorialspoint.com/computer_programming/index.htm

[Introduction to web Technologies could be found at MaharaTech](http://bit.ly/2OGcLix)
http://bit.ly/2OGcLix

[Udacity](https://classroom.udacity.com/courses/ud001)
https://classroom.udacity.com/courses/ud001

[Introduction to Database could be found at MaharaTech](https://maharatech.gov.eg/enrol/index.php?id=740)
https://maharatech.gov.eg/enrol/index.php?id=740

[Khan Academy](https://www.khanacademy.org/computing/computer-programming/sql)
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APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.